

Mirko Di Maggio

AI Engineer



About me

I'm an AI Engineer driven by curiosity and a passion for turning ideas into reality. I specialize in building intelligent systems that don't just work—they make a difference. From machine learning models to scalable applications and seamless API integrations, I enjoy bridging the gap between innovative technology and real-world impact. Problem-solving is where I thrive, especially when it involves blending creativity and code to craft solutions that are both elegant and effective.

Personal info

Mirko Di Maggio
Nationality: Italian
Born: 2001

Core Skills

Multi-Agent Systems • LLMs & RAG Orchestration • Edge AI
• Model Quantization (INT8/FP16) • Computer Vision • Self-Supervised Learning (SSL)
Scalable AI Architectures
• MLOps (Docker/K8s) • API Design (FastAPI/Node.js)
Machine Learning • AI API Integration • Natural Language Processing (NLP)
• Scalable Web Systems
Deployment with Docker/Kubernetes
• Database Management (SQL/NoSQL) • Cloud Solutions

Interests

Basketball • Robotics
• AI • Technologies



PROFESSIONAL EXPERIENCE

Sept. 2025

Hackathon Participant

ITALIAN TECH WEEK • Turin, Italy 📍

Developed "Spaice", an Edge AI gesture interface for museums on Arduino Nicla Vision. Designed a dual-model pipeline (DETR detector + CNN classifier) using the Focoos framework. Optimized for real-time execution via INT8 quantization and QSPI/XIP weight storage. Built a Python-WebSocket relay to stream interaction data to a Unity 3D engine.



Sept. 2025

Multi-Agent Systems Developer

SWISS [AI] WEEKS - UBS CHALLENGE • Zurich, Switzerland 📍

Architected an AI-driven system to automate financial workflows at the ETH AI Center. Developed a dual-classifier architecture using GPT and Apertus-8B (ETH/CSCS). Engineered a "Disagreement Gate" mechanism with a "Judge" model for automated quality control and conflict resolution. Collaborated with UBS engineers and ETH researchers to optimize escalation logic.



Aug. 2025

Selected Participant

SILICON VALLEY STUDY TOUR • San Francisco, USA 📍

Intensive immersion in the Bay Area innovation ecosystem. Visited tech giants (Google, Apple, LinkedIn, Zoom) and academic hubs (Stanford, UC Berkeley). Engaged in mentorship sessions with industry leaders (L. Degioanni, V. Viarengo) on AI/ML execution, product management, and startup scaling. Refined pitching skills and built a high-level international network.



May 2025

Software Engineer – Hackathon Participant

HACKUPC 2025 • Barcelona 📍

Ranked 4th out of 150+ international teams in the Skyscanner challenge. Co-developed OnlyFly, an AI-powered home-exchange platform enabling travelers to swap homes and eliminate accommodation costs. Built a fully functional MVP in 36 hours with features like smart AI matching, secure payments, and sustainability insights.



Feb. 2025

AI Developer

7TH PLACE - GENAI HACKATHON • Remote 📍

Placed 7th out of 4,500 participants from 80 countries. Co-developed PostGenius, an AI-powered app for generating multimedia content (text, images, memes, videos) from news articles using Retrieval-Augmented Generation (RAG) techniques. Contributed to system architecture, backend development with FastAPI, and API integration (OpenAI, Vectara, RunwayML). The app streamlines social media content creation for platforms like Twitter and Instagram.



2024

AI App Developer

HACK THIS FALL • Remote 📍

Built a scalable AI application leveraging Gemini API for multimedia content processing. Optimized front-end UI for seamless user interaction and integrated backend API services.



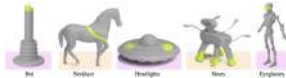
PROFESSIONAL EXPERIENCE (CONTINUED)

Oct. 2023 – Dec. 2023	Front-End Developer Evo HUNT · Torino 📍 Developed and customized back-office web applications using React.js and JavaScript. Collaborated with teams to integrate dynamic data management solutions.	
Mar. 2023 – Sep. 2023	Tester Debugger TODUBA · Torino 📍 Conducted functional, security, and usability testing on iOS/Android platforms for a ticketing app. Provided actionable insights to improve performance.	

PUBLICATIONS

2025	Research Paper (In Progress) SELF-SUPERVISED LEARNING FOR RCC CLASSIFICATION · Politecnico di Torino 📍 Authoring a research paper on Renal Cell Carcinoma (RCC) subtype classification. Benchmarked SSL architectures (MoCo v3, Dino v3, I-JEPA, iBot) on histopathological slides to reduce dependency on annotated data. Optimized feature extraction and patient-level inference strategies.	
------	--	---

PERSONAL PROJECTS

2025	Affordance Highlighting Project 3D NEURAL AFFORDANCE DETECTION · GitHub 📍 Explored neural fields for 3D affordance detection using CLIP-based supervision. Adapted 3D Highlighter to point clouds and evaluated the system on the AffordanceNet benchmark using mIOU. Focus: neural fields · CLIP · differentiable rendering.	
------	--	---

EDUCATION

2024–2025	Master of Engineering ARTIFICIAL INTELLIGENCE AND DATA ANALYTICS · Politecnico di Torino 🏛️ MEng	
2020–2024	Bachelor's Degree COMPUTER ENGINEERING · Politecnico di Torino 🏛️ BSc	

PROGRAMMING

Python	
PyTorch / HF	
FastAPI / Node	
React / JS	
Docker / K8s	
Edge AI / C++	
SQL / NoSQL	
Rust	
Git / WebSockets	

CERTIFICATES & GRANTS

2023	IELTS Certification - B2 Level
------	--------------------------------

LANGUAGES

Italian	C2	mother tongue
English	B2	